

Exercise n. 2

Write a numerical code to solve the following partial differential equation, subject to the prescribed initial and boundary conditions, over the spatial and time intervals , $x \in [0, L]$ and $t \in [0, T]$:

$$\frac{\partial \phi(x, t)}{\partial t} = -c \frac{\partial \phi(x, t)}{\partial x} + \nu \frac{\partial^2 \phi(x, t)}{\partial x^2}$$

where $c > 0$ is the constant convection velocity and $\nu \geq 0$ is the constant diffusion coefficient.

The boundary conditions should be periodic, $\phi(0, t) = \phi(L, t) \forall t \in [0, T]$.

Assume $L = 1$, while T is given below and varies depending on the case considered.

The problem should be solved by using a **three-stage Runge-Kutta time integration method**.

The first-order spatial derivative (convection term) should be discretized using one of the following finite difference schemes (implement all of them):

- I. **First-order forward (FW1)**
- II. **First-order backward (BW1)**
- III. **Second-order centered (CS2)**

The second-order spatial derivative (diffusion term) should be discretized using the following finite difference scheme:

- A. **Fourth-order centered (CS4, see the course notes).**

Perform the following tasks for different cases, by varying the parameters c and ν :

1) Pure convection ($\nu = 0$, $c = 0.1$)

- A. Consider the following Gaussian initial condition,

$$\phi(x, 0) = e^{-\frac{(x-x_0)^2}{2\sigma^2}} \text{ with } x_0 = L/2 \text{ and } \sigma = 0.1,$$

together with periodic boundary conditions. Solve the problem over the time interval $t \in [0, 1]$, using a time-step $\Delta t = 10^{-2}$ and a uniform spatial discretization with $N_x = 100$ grid points. Compute the numerical solution using the spatial scheme III (CS2). **Evaluate the CFL number and plot, in a single figure, the numerical and exact solutions at different equally spaced time instants.**

B. Consider the following harmonic initial condition,

$$\phi(x,0) = \sin\left(\frac{2\pi x}{L}\right),$$

together with periodic boundary conditions. Solve the problem over the time interval $t \in [0,100]$, using a time-step $\Delta t = 0.25$ and a uniform spatial discretization with $N_x = 50$ grid points. Compute the numerical solution using spatial scheme I (FW1), scheme II (BW1), and scheme III (CS2). Evaluate the CFL number and plot, in three separate figures, the numerical and exact solutions at different equally spaced time instants, using one figure for each spatial scheme. Pay particular attention to the behavior of the solution obtained with the different discretizations of the first-order derivative, and discuss the observed differences.

2) Pure diffusion ($\nu = 0.01$, $c = 0$)

A. Consider the following Gaussian initial condition,

$$\phi(x,0) = e^{-\frac{(x-x_0)^2}{2\sigma^2}} \text{ with } x_0 = L/2 \text{ and } \sigma = 0.1,$$

together with periodic boundary conditions. Solve the problem over the time interval $t \in [0,3]$, using a time-step $\Delta t = 10^{-3}$ and a uniform spatial discretization with $N_x = 100$ grid points. Compute the numerical solution using spatial scheme A (CS4). Evaluate the diffusion number, D , **and plot, in a single figure, the numerical and exact solutions at different time instants.**

3) Convezione-diffusione ($\nu = 0.01$, $c = 0.2$)

A. Choose either the Gaussian or the harmonic initial condition and solve the problem with the usual periodic boundary conditions. Select an arbitrary combination of spatial discretization schemes among those introduced above or presented in the course notes. Then choose a suitable time step and grid spacing, checking the corresponding CFL and diffusion numbers, D . Finally, discuss the obtained results and present, in one or more figures, the numerical and exact solutions at different time instants.

OBS: The report can be prepared individually or in groups of up to three students. Conciseness, clarity of presentation, and the quality of figures are appreciated.

HINT: The exact solution of the problem corresponding to a Gaussian initial condition with periodic boundary conditions is given by:

$$\phi(x,t) = \frac{\sigma}{\sqrt{\sigma^2 + 2\nu t}} \exp\left(-\frac{(x - x_0 - ct)^2}{2(\sigma^2 + 2\nu t)}\right),$$

whereas for the harmonic signal by:

$$\phi(x,t) = \exp\left(-\nu \left(\frac{2\pi}{L}\right)^2 t\right) \sin\left[\frac{2\pi}{L}(x - ct)\right].$$